Joint Scheduling and Workforce Allocation for Aircraft Pulsating Assembly Lines via Heterogeneous Graph Reinforcement Learning

Xiao Lu, Jiakang Ai, Fei Qiao, Juan Liu, Lingtao Chen, and Yumin Ma

***Abstract*—Operation planning in an aircraft pulsating assembly line (APAL) involves tightly coupled constraints, including technological precedence, station concurrency capacities, multi-skilled workers, and team-based execution. Operation-to-station assignment and worker allocation interact intricately, while combinatorial team selection drastically expands the action space. To address these challenges, this paper proposes HGP-PPO, a reinforcement learning framework combining heterogeneous graph neural networks with an autoregressive pointer decoder. Cyclic operation template construction is formulated as a sequential Markov decision process over a heterogeneous graph with Operation, Station, Worker, and Skill nodes. A hierarchical operation–station–worker autoregressive decoder sequentially selects operations, stations, and qualified worker teams, enforcing feasibility via dynamic masking and an allocation engine while endogenously coordinating worker–station assignments. By augmenting state representations with baseline and disturbance features, the policy adapts to stability-aware rescheduling under material delivery delays. Extensive evaluations on four real factory instances (283 to 3,182 physical operations) demonstrate that a single locked policy generates 100% physically valid schedules without fine-tuning, reducing average cycle span by 2.22% over L2D-PPO-APAL and achieving the lowest average makespan of 902.40 h. Across 36 locked material delay scenarios, the rescheduling model maintains full feasibility, reducing composite recovery scores by 7.00% and 3.01% against L2D-PPO-APAL and StabilityAwareRepair, while completing full-line plan repair in approximately two minutes on the largest instance.**

***Index Terms*—Aircraft pulsating assembly line, joint scheduling, workforce allocation, heterogeneous graph neural network, proximal policy optimization, stability-aware rescheduling.**

I. Introduction

AIRCRAFT final assembly is a critical final stage of aircraft manufacturing, encompassing operations such as major component mating, piping installation, engine installation, avionics integration, and functional testing. Unlike conventional shop scheduling or automated mass-production lines, aircraft final assembly deals with large, highly customized products characterized by extensive process chains, dense precedence dependencies, long operation durations, restricted working space, and collaboration among multiple trades [1]–[3]. To improve production flow and establish a regular production rhythm, aircraft manufacturers widely adopt aircraft pulsating assembly lines (APALs). An APAL partitions the final assembly shop floor into several dedicated physical stations, through which aircraft under assembly advance synchronously in a pulsating cadence. Within each pulsation cycle, each aircraft remains stationary at its assigned station while a designated set of operations is executed. Upon cycle completion, all aircraft along the line advance synchronously to their respective downstream stations until final roll-out [4].

This synchronous advancement mechanism allows local operation delays to affect the timing of line-wide station transfers. Line advancement is strictly dictated by the bottleneck station with the latest completion time. Delays caused by uneven task distribution, shortages of qualified workers, or station concurrency limits can postpone synchronous advancement when they extend the latest station completion time, requiring aircraft at stations that have already finished to wait. Consequently, jointly coordinating operation-to-station assignment, within-station sequencing, and resource allocation is essential to efficient and stable APAL operation [2], [5], [6].

In practical shop-floor settings, APAL scheduling involves three interdependent decisions for each operation: its execution sequence, its assigned physical station, and a specific collaborative worker team that satisfies the required skill qualifications and team size. Pre-assigning workers to dedicated stations before task distribution is determined can limit the ability to match workforce capabilities with evolving station workloads. The proposed method therefore determines worker–station assignments endogenously during schedule construction as workers are first selected for station-specific operations. This mechanism gives team selections lasting downstream consequences: each assignment not only determines an operation's execution duration and immediate resource consumption, but also binds previously unassigned workers to a specific station, thereby restricting the labor pool available to subsequent operations. Consequently, the execution efficiency of the current operation and the anticipated skill demands of downstream stations must be coordinated within the same unified decision process.

This joint decision problem presents several challenges:

1) Operations, stations, workers, and skills exhibit heterogeneous multi-relational topologies, operating within an environment characterized by rapid state transitions and dynamic resource calendars;
2) Selecting an optimal team from a multiskilled labor pool results in a combinatorial team-selection space of size

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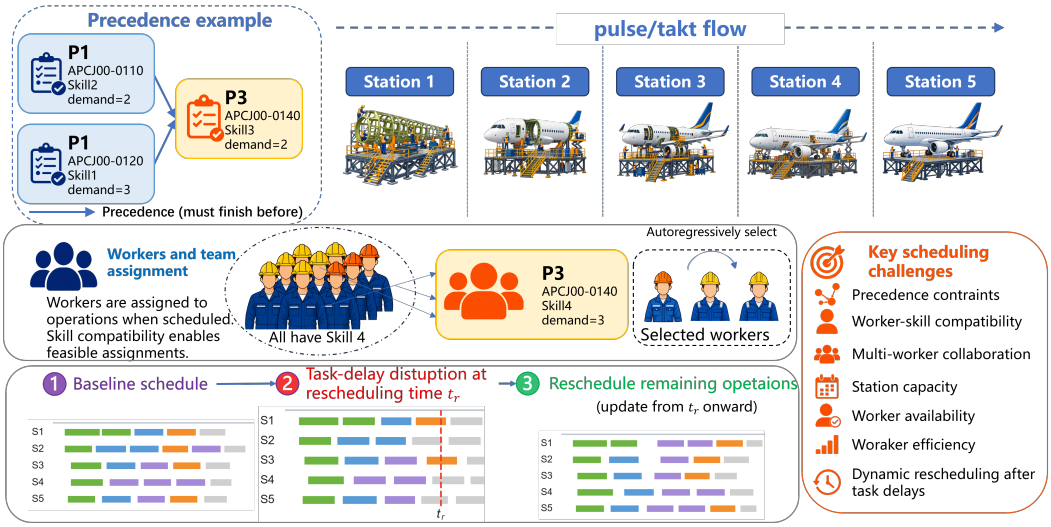


Fig. 1. Illustration of cyclic operation scheduling and stability-aware rescheduling for an aircraft pulsating assembly line. Aircraft advance through five physical stations in a pulsating rhythm. The cyclic operation template jointly determines the precedence-related execution relationships among operations, their assigned stations, and specific worker teams satisfying skill and headcount requirements. Following a material delay, the system locks committed entries at the schedule freeze boundary τ^r, inherits worker bindings, and performs stability-aware rescheduling of the remaining operations.

$\binom{n_i}{q_i}$ for each operation;

3) Each intermediate decision changes the feasible choices available to subsequent decisions;
4) When shop-floor disturbances such as material arrival delays occur, rescheduling must preserve frozen schedule entries and existing worker–station bindings while repairing subsequent cyclic templates to restore feasibility and balance completion span against deviations from the baseline schedule [7], [8].

Fig. 1 illustrates the physical setting, entity interactions, and the plan repair process under material delay disturbances.

To address the coupling between operation scheduling and workforce allocation in aircraft pulsating assembly lines, this paper proposes the HGP-PPO joint scheduling framework. The main contributions are as follows:

1) **Joint scheduling model with endogenous workforce allocation:** An APAL cyclic operation template model is formulated to incorporate precedence constraints, station concurrency limits, and collaborative operations performed by multi-skilled workers. The model integrates operation-to-station assignment, selection of specific worker teams, and worker–station assignment into a unified sequential construction process. Persistent station bindings established upon each worker's first assignment explicitly capture how current team choices affect subsequent worker availability, enabling the coordinated determination of operation schedules and station workforce allocation.
2) **Heterogeneous graph representation of process dependencies and skill supply–demand relationships:** A heterogeneous graph comprising operation, station, worker, and skill nodes jointly represents process dependencies, skill supply–demand relationships, and dynamic assignments. Relation-specific attention encoding provides a shared state representation for operation-to-station assignment and team formation, enabling the policy to make joint decisions informed by both task structures and workforce states.
3) **Autoregressive joint decoding incorporating team states and skill scarcity:** A conditional operation–station–worker pointer mechanism decomposes the combinatorial selection of fixed-size teams into sequential member selections. Candidate workers are scored conditionally using partial team representations, tentative completion-time features, and the marginal scarcity of their additional skills. This mechanism avoids explicit enumeration of complete teams and uses dynamic masking and forward time allocation to enforce the scheduling constraints.

The framework is evaluated for both initial scheduling and rescheduling under material readiness delays. For initial scheduling, a single policy is held fixed and applied without fine-tuning to four test instances containing 283 to 3,182 physical operations. Comparisons with workforce pre-assignment, alternative graph representations, and decision architectures assess the role of the overall design. The framework is then adapted to stability-oriented cyclic template rescheduling by incorporating baseline schedule and disturbance features and continuing training, while preserving frozen operations and existing worker–station bindings. Evaluation across 36 paired disturbance scenarios considers schedule feasibility, completion efficiency, deviations from the baseline schedule, and online solution time. Together, these experiments assess cross-scale applicability within the investigated APAL data family and the framework's applicability to both initial template construction and repair under material readiness delays.

The remainder of this paper is organized as follows: Section II reviews related work. Section III details the APAL scheduling and rescheduling models. Section IV presents the HGP-PPO methodology. Section V provides experimental results and ablations. Section VI concludes the paper with discussion on limitations and future avenues.

II. Related Work

A. Aircraft Assembly Scheduling and Workforce Allocation

Assembly line balancing and scheduling, spanning ALBP, SALBP, and GALBP, provide classical foundations for workload assignment across stations [5], [6], [9]. In modern aerospace manufacturing, complex structural assembly operations have motivated the incorporation of multi-resource constraints. Bao *et al.* [2], [10] addressed aircraft final assembly line balancing with resource constraints using staged heuristic searches. Fang *et al.* [11] and Wang *et al.* [3] examined flexible cross-station workforce allocation. Liu *et al.* [12] and Wen *et al.* [13] studied multiskilled worker assignments and skill-grade matching during intra-station dispatching. While these studies underscore the decisive role of multi-resource allocation, most existing approaches treat station assignments as predetermined or restrict decision-making to aggregate workforce sizing. In contrast, this paper tackles a global setting where stations and worker teams are resolved jointly from globally unbound pools in a unified decision flow.

B. Graph Reinforcement Learning for Production Scheduling

Graph neural networks (GNNs) learn structural representations of combinatorial optimization problems through shared message-passing functions [14]. In job-shop (JSSP) and flexible job-shop scheduling (FJSP), graph-based reinforcement learning methods encode operation dependencies and machine constraints to learn dispatching policies [15]–[17]. In aerospace assembly, our preliminary work [18] showed the promise of graph-enhanced reinforcement learning for pulsating lines. Wen *et al.* [19] paired graph isomorphism networks (GIN) with PPO to address skill-level assembly scheduling. Nevertheless, standard RL action spaces typically focus on selecting a single operation or an operation–machine pair. To handle collaborative teams and dynamic station binding, the proposed framework represents process dependencies and skill supply–demand relationships through a heterogeneous graph and constructs teams using an operation–station–worker autoregressive decoder. Partial team states and skill-scarcity features inform successive member selections, while dynamic masks account for worker qualifications and station bindings.

C. Assembly Line Rescheduling and Plan Stability

In dynamic manufacturing, predictive–reactive scheduling provides a structured paradigm wherein baseline plans are formulated and subsequently revised upon disturbance arrivals [7], [20]. Regarding supply chain disruptions, Zhu *et al.* [21] adjusted unstarted operations for moving lines. Hu *et al.* [8] developed a two-stage robust strategy under processing-time variations, while Chen *et al.* [22] utilized multi-agent RL for dynamic dispatching under material delays. In APALs, rescheduling must account for synchronous line advancement. This study focuses on plan-level cyclic template rebalancing under material readiness delays, preserving frozen operations and existing worker–station bindings. Baseline schedule and disturbance features are incorporated into the joint decision framework, which is further trained to balance completion efficiency against deviations in operation start times, station assignments, and team composition.

III. APAL Joint Scheduling and Rescheduling Problem Formulation

A. System Description, Assumptions, and Notation

Consider an APAL comprising five fixed physical stations $\mathcal{M} = \{1, \ldots, 5\}$. The assembly process is represented by a directed acyclic graph (DAG) $G = (\mathcal{O}, \mathcal{E})$, where $\mathcal{O} = \mathcal{O}^+ \cup \mathcal{O}^0$. Here, $\mathcal{O}^+$ denotes physical operations with positive processing times, and $\mathcal{O}^0$ denotes non-executable process-structure nodes (having zero duration and retaining only process hierarchies, hereafter referred to as structural nodes). Structural nodes consume no physical resources. Physical precedence relations are derived by bypassing structural nodes while preserving the dependencies between physical operations, forming the physical operation DAG $G^+ = (\mathcal{O}^+, \mathcal{E}^+)$. The immediate physical predecessors of operation i are denoted by $\mathcal{P}(i)$.

The system includes a pool of multiskilled workers $\mathcal{W}$ and a set of five skill types $\mathcal{K}$. Each physical operation $i \in \mathcal{O}^+$ has a standard processing time $p_i^{\mathrm{std}} > 0$, requires a single skill $k_i \in \mathcal{K}$, and demands a fixed collaborative team size $q_i \in \mathbb{N}^+$. Each worker $w \in \mathcal{W}$ has a binary skill qualification vector $\mathbf{H}_w = (H_{wk})_{k \in \mathcal{K}}$, where $H_{wk} \in \{0, 1\}$, and a baseline efficiency coefficient $\eta_w > 0$. Except for initial positioning tasks, operation stations are not predetermined.

The model adopts the following operating assumptions:

1) *Non-preemptive operations:* Once an operation starts, it proceeds without interruption, and its assigned station and worker team remain unchanged until completion.
2) *Transfer and setup times:* Operation transfer and setup times are included in the standard processing times provided by the factory.
3) *Resource availability:* Workers and stations remain operational throughout the planning horizon, with no unexpected equipment failures or worker absences. Resource use remains subject to worker exclusivity and station concurrency limits.
4) *Worker–station binding:* All workers are initially unassigned to stations. Each worker's station assignment is established upon their first task assignment and remains fixed throughout the initial schedule. Workers therefore cannot serve operations at different stations within that schedule. These station assignments are retained during subsequent rescheduling.

Table I outlines the mathematical notation used throughout this paper.

B. Initial APAL Cyclic Scheduling Model

The processing duration of operation i depends on the efficiencies of its assigned workers and a team coordination discount factor $\rho = 0.95$:

$$D_i(W_i) = \frac{p_i^{\mathrm{std}} q_i}{\rho^{q_i - 1} \sum_{w \in W_i} \eta_w}. \tag{1}$$

TABLE I
MAIN MATHEMATICAL NOTATION

Symbol	Physical Meaning
$G^+ = (\mathcal{O}^+, \mathcal{E}^+)$	Physical operation DAG and its precedence edges
$\mathcal{M}, \mathcal{W}, \mathcal{K}$	Physical stations ($\lvert\mathcal{M}\rvert = 5$), worker pool, skill types ($\lvert\mathcal{K}\rvert = 5$)
$p_i^{\mathrm{std}}, q_i, k_i$	Standard duration, required team size, and skill type of operation i
H_{wk}, η_w	Binary skill indicator and baseline efficiency of worker w
c_m	Concurrency capacity (maximum simultaneous tasks) of station m
x_{im}, y_{iw}	Binary decision variables for station assignment and worker assignment
m_i, W_i	Assigned physical station and specific team for operation i ($\lvert W_i\rvert = q_i$)
B_i, C_i	Start time and completion time of operation i on template timeline
ℓ_w	Station binding of worker w; $\ell_w = \varnothing$ indicates an unbound worker
$\pi^0, T^0_{\mathrm{ref}}$	Baseline template and reference cycle span $T^0_{\mathrm{ref}} := H^0 = C_{\max}(\pi^0)$
τ^r, r_i	Rescheduling freeze boundary and updated material release time
$\mathcal{O}^{\mathrm{frz}}, \mathcal{O}^{\mathrm{mov}}$	Frozen operations ($B_i^0 \le \tau^r$) and reschedulable operations ($B_i^0 > \tau^r$)
$H, C_{\max}, S$	Cyclic template span, template makespan, and composite recovery objective

The optimization objective minimizes the template cycle span $H(\pi) := C_{\max}(\pi)$:

$$\min_{\pi} \; C_{\max}(\pi) = \max_{m\in\mathcal{M}} C_m^{\mathrm{end}}(\pi) = \max_{i\in\mathcal{O}^+} C_i, \tag{2}$$

where $C_m^{\mathrm{end}}(\pi) = \max_{\{i\in\mathcal{O}^+ \mid m_i=m\}} C_i$ represents the latest completion time at station m. Once the baseline schedule is fixed, its makespan defines the reference cycle span $T^0_{\mathrm{ref}} := H^0 = C_{\max}(\pi^0)$, which remains unchanged during rescheduling. Station labor load $L_m = \sum_{m_i=m} q_i D_i(W_i)$ serves as an intermediate diagnostic and reward-shaping signal.

The schedule must satisfy the following assignment, precedence, resource, and temporal constraints:

1) **Unique assignment and station monotonicity**:

$$\sum_{m\in\mathcal{M}} x_{im} = 1, \quad m_j \ge m_i, \quad \forall (i,j) \in \mathcal{E}^+. \tag{3}$$

2) **Intra-template temporal precedence**:

$$B_j \ge C_i, \quad \forall (i,j) \in \mathcal{E}^+. \tag{4}$$

3) **Team size, skill qualifications, and station exclusivity**:

$$\sum_{w\in\mathcal{W}} y_{iw} = q_i, \quad y_{iw} \le H_{wk_i}, \quad \forall i \in \mathcal{O}^+, w \in \mathcal{W}; \tag{5}$$

$$y_{iw} = y_{jw} = 1 \Longrightarrow m_i = m_j, \quad \forall i,j \in \mathcal{O}^+, w \in \mathcal{W}. \tag{6}$$

4) **Temporal and spatial concurrency capacity**:

$$\sum_{i\in\mathcal{O}^+} y_{iw}\mathbb{I}(B_i \le t < C_i) \le 1, \quad \forall w \in \mathcal{W}, t \ge 0; \tag{7}$$

$$\sum_{i\in\mathcal{O}^+} x_{im}\mathbb{I}(B_i \le t < C_i) \le c_m, \quad \forall m \in \mathcal{M}, t \ge 0. \tag{8}$$

5) **Forward allocation lower bounds**:

$$B_i \ge \max\left\{ c_t, \max_{j\in\mathcal{P}(i)} C_j, \max_{w\in W_i} a_w \right\}, \tag{9}$$

where c_t is the current environment time clock and a_w is the earliest available time of worker w.

C. Steady-State Cyclic Unfolding Across Aircraft

Proposition 1 (Physical Feasibility of Cyclic Unfolding): Given a feasible cyclic template $\pi = (m, W, B, C)$ satisfying all constraints in Section III-B with span $H = C_{\max}$, let aircraft index be $k = 0, 1, 2, \ldots$. For aircraft k, the execution interval of operation i on the shop-floor timeline is

$$B_{i,k}^{\mathrm{phys}} = (k+m_i-1)H + B_i, \quad C_{i,k}^{\mathrm{phys}} = (k+m_i-1)H + C_i. \tag{10}$$

Under these assumptions, the unfolded schedule preserves aircraft occupancy exclusivity, precedence feasibility, worker exclusivity, and station concurrency limits.

Proof: Aircraft k resides at station m during $[(k+m-1)H, (k+m)H)$, and all its operations assigned to that station are contained within this window. Consecutive aircraft at the same station occupy non-overlapping intervals, ensuring spatial exclusivity. For any $(i,j) \in \mathcal{E}^+$, the physical time difference satisfies:

$$B_{j,k}^{\mathrm{phys}} - C_{i,k}^{\mathrm{phys}} = (m_j - m_i)H + (B_j - C_i) \ge 0. \tag{11}$$

Since $m_j \ge m_i$ and $B_j \ge C_i$, physical precedence is naturally preserved. For cross-station dependencies where $m_j > m_i$, $(m_j - m_i)H \ge H$, meaning predecessor tasks were completed in earlier cycles. Imposing $B_j \ge C_i$ within a single template thus serves as a structured sufficient condition for physically executable multi-aircraft unfolding. ■

D. Stability-Aware Template Rescheduling Under Material Delays

Predictive–reactive template repair setting: The baseline schedule π^0 defines a cyclic operation template for repeated production. Following a material readiness delay, the template is revised at the plan level for subsequent production while preserving frozen operations and existing worker–station assignments. Updated material release times are assumed known when rescheduling is triggered.

Let $\mathcal{O}^{\mathrm{del}} \subseteq \mathcal{O}^{\mathrm{mov}}$ denote the set of operations affected by material readiness delays. For each $i \in \mathcal{O}^{\mathrm{del}}$, the updated release time is $r_i = B_i^0 + \Delta_i$, where $\Delta_i > 0$. At the schedule freeze boundary τ^r, the physical operations are partitioned into frozen set $\mathcal{O}^{\mathrm{frz}} = \{i \in \mathcal{O}^+ \mid B_i^0 \le \tau^r\}$ and reschedulable set $\mathcal{O}^{\mathrm{mov}} = \{i \in \mathcal{O}^+ \mid B_i^0 > \tau^r\}$.

1) **Frozen commitments**: Committed operations strictly inherit baseline assignments:

$$(m_i^r, W_i^r, B_i^r, C_i^r) = (m_i^0, W_i^0, B_i^0, C_i^0), \quad \forall i \in \mathcal{O}^{\mathrm{frz}}. \tag{12}$$

2) **Worker binding inheritance**: Workers chosen in the baseline plan retain their station bindings ($\ell_w^r = m$). Only previously unused workers remain unbound.

3) **Reschedulable bounds**: Reschedulable operations must satisfy $B_i^r \geq \max\{\tau^r, r_i\}, \forall i \in \mathcal{O}^{\mathrm{mov}}$.

Defining $N_{\mathrm{mov}} = |\mathcal{O}^{\mathrm{mov}}|$, baseline overrun $\Delta_{\mathrm{ref}}^r = \max\{0, C_{\max}^r - T_{\mathrm{ref}}^0\}$, start-time deviation $\bar{D}_{\mathrm{start}}^r = \frac{1}{N_{\mathrm{mov}}} \sum |B_i^r - B_i^0|$, station change rate $R_{\mathrm{sta}}^r = \frac{1}{N_{\mathrm{mov}}} \sum \mathbb{I}(m_i^r \neq m_i^0)$, and team change rate $R_{\mathrm{team}}^r = \frac{1}{N_{\mathrm{mov}}} \sum \mathbb{I}(W_i^r \neq W_i^0)$, the composite stability objective is formulated as:

$$\min_{\pi^r \in \Pi_{\mathrm{feas}}^r} S(\pi^r) = 0.20 \frac{C_{\max}^r}{T_{\mathrm{ref}}^0} + 3 \frac{\Delta_{\mathrm{ref}}^r}{T_{\mathrm{ref}}^0} + 4 \frac{\bar{D}_{\mathrm{start}}^r}{T_{\mathrm{ref}}^0} + 4R_{\mathrm{sta}}^r + 0.3R_{\mathrm{team}}^r. \tag{13}$$

IV. HGP-PPO Method

A. Unified Event-Driven MDP Formulation

Initial scheduling and stability-aware rescheduling are represented within a unified event-driven MDP framework:

$$\mathcal{M}^z = (\mathcal{S}^z, \mathcal{A}, \mathcal{P}, \mathcal{R}^z, \gamma), \qquad z \in \{\mathrm{init}, \mathrm{res}\}.$$

1) **State representation**: State s_t at decision step t is encoded as a four-entity heterogeneous graph G_t capturing process dependencies, station loads, and worker availability calendars. The index t denotes a scheduling decision step, whereas c_t denotes the current environment time.
2) **Hierarchical action decomposition**: The discrete joint action $a_t = (i_t, m_t, W_t)$ ($|W_t| = q_{i_t}$) is autoregressively decomposed into an ordered trajectory $\widetilde{a}_t = (i_t, m_t, \mathbf{w}_t)$, mapped to unordered set W_t upon team completion.
3) **Dynamic masking and event-driven state transition**: Hard constraints are enforced at each decision tier by masking invalid candidates with $-\infty$. Given the current partial schedule and the selected joint action, the forward allocation engine deterministically assigns the earliest feasible start time B_i subject to predecessor completion, worker availability, material release times, and station concurrency limits. It then sets $C_i = B_i + D_i(W_i)$ and updates the resource calendars. When no scheduling action is currently feasible, the environment clock advances to the next event that may enable further scheduling. If no such event exists while operations remain unscheduled, the episode terminates in deadlock.
4) **Stage-specific rewards**: Initial scheduling uses differential makespan rewards and changes in the standard deviation of station labor loads: $\widetilde{r}_t^{\mathrm{init}} = -(\widehat{C}_{t+1} - \widehat{C}_t)/C_{\mathrm{ref}} - 0.5(\sigma_{L,t+1} - \sigma_{L,t})/L_{\mathrm{ref}}$; rescheduling directly adopts the negative composite score difference: $\widetilde{r}_t^{\mathrm{res}} = -[\widehat{S}(s_{t+1}) - \widehat{S}(s_t)]$.

Fig. 2 presents the architectural layout of HGP-PPO.

B. Heterogeneous Graph Representation and Global Encoding

At decision step t, the state is represented by a heterogeneous graph $G_t = (\mathcal{V}_t, \mathcal{E}_t)$, where $\mathcal{V}_t = \mathcal{O} \cup \mathcal{M} \cup \mathcal{W} \cup \mathcal{K}$. The Operation node set $\mathcal{O} = \mathcal{O}^+ \cup \mathcal{O}^0$ includes physical operations and auxiliary structural nodes retained from the source process graph. Structural nodes have zero duration and no physical resource requirements; they are retained for graph encoding but excluded from scheduling actions. Tables II and III summarize the node features and relation types, respectively.

TABLE II
Input State Features for the Four Node Types

Node Type	State Features
Operation	Normalized duration $\widetilde{p}_i$, status, 5-dim skill requirement, team size q_i, release time; rescheduling adds baseline start, station, team size, and freeze/delay flags
Station	Cumulative labor load, bound/unbound worker ratios, slot waiting time, available concurrency slots
Worker	Baseline efficiency η_w, 5-dim skills, waiting time, idle flag, station binding ℓ_w
Skill	Category index, headcounts, available capacity, and mean efficiency of qualified workers

Note: Operation-specific resource requirements apply to physical operations.

TABLE III
Heterogeneous Graph Relations and Dynamics

Relation	Edge Direction	Dynamics	Physical Meaning
precedes	Operation → Operation	Static	Process precedence dependency
assigned-to	Operation → Station	Dynamic	Task station assignment
has-task	Station → Operation	Dynamic	Station-hosted operations
done-by	Operation → Worker	Dynamic	Assigned team members
has-skill	Worker → Skill	Fixed	Worker skill qualification
provided-by	Skill → Worker	Fixed	Skill-holding workforce
requires	Operation → Skill	Static	Operation skill requirement
required-by	Skill → Operation	Static	Operations requiring skill

Skill nodes bridge operation skill requirements with worker skill competencies, enabling operations sharing identical skill demands and workers possessing multiple qualifications to engage in message passing through explicit relational edges. In conjunction with process precedence and dynamic assignment relations, this representation establishes a unified graph context for the policy to evaluate immediate task demands and concurrent labor competition. Raw features are projected into a d-dimensional space $\mathbf{h}_v^{(0)} = f_X(\mathbf{x}_v)$. Message passing is executed across $L = 5$ relation-specific GATv2 layers [23]. For edge $u \to v$ under relation r, head h:

$$e_{uv}^{(r,\ell,h)} = \mathbf{a}_{r,h}^{(\ell)\top} \mathrm{LeakyReLU}\left(\mathbf{W}_{s,r,h}^{(\ell)} \mathbf{h}_u^{(\ell-1)} + \mathbf{W}_{t,r,h}^{(\ell)} \mathbf{h}_v^{(\ell-1)}\right), \tag{14}$$

$$\alpha_{uv}^{(r,\ell,h)} = \frac{\exp(e_{uv}^{(r,\ell,h)})}{\sum_{z \in \mathcal{N}_r(v)} \exp(e_{zv}^{(r,\ell,h)})}. \tag{15}$$

Aggregating multi-head messages $\mathbf{m}_v^{(r,\ell)} = \frac{1}{H} \sum_{h=1}^{H} \sum_{u \in \mathcal{N}_r(v)} \alpha_{uv}^{(r,\ell,h)} \mathbf{W}_{s,r,h}^{(\ell)} \mathbf{h}_u^{(\ell-1)}$:

$$\mathbf{h}_v^{(\ell)} = \mathbf{h}_v^{(\ell-1)} + \mathrm{LeakyReLU}\left(\sum_{r \in \mathcal{R}_{\mathrm{in}}(v)} \mathbf{m}_v^{(r,\ell)}\right). \tag{16}$$

Node-type attention pooling produces a fixed-dimensional, permutation-invariant global context $\mathbf{g}_t = [\mathbf{g}_M \,\|\, \mathbf{g}_O \,\|\, \mathbf{g}_W] \in \mathbb{R}^{3d}$:

$$\beta_v^X = \frac{\exp(a_X(\mathbf{h}_v))}{\sum_{u \in \mathcal{V}^X} \exp(a_X(\mathbf{h}_u))}, \quad \mathbf{g}_X = \sum_{v \in \mathcal{V}^X} \beta_v^X \mathbf{h}_v. \tag{17}$$

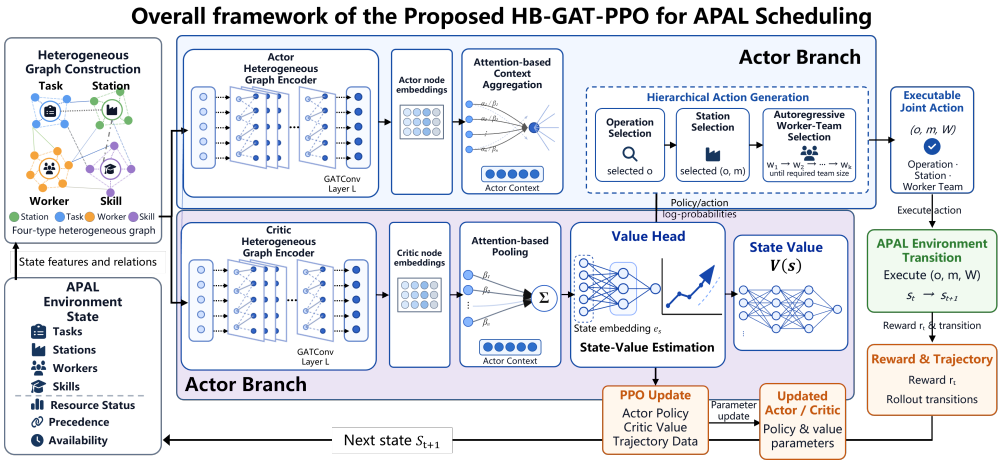


Fig. 2. Overall HGP-PPO framework. The APAL environment state is first converted into a heterogeneous graph with Operation, Station, Worker, and Skill nodes. The Actor and Critic use independent heterogeneous graph encoders. After attention-based context aggregation, the Actor sequentially selects an Operation, a Station, and a Worker team of the required size to form an executable joint action (o, m, W). The Critic aggregates the graph state and estimates the state value $V(s)$. Executing the joint action produces the next state, reward, and trajectory, which PPO uses to update the policy and value networks.

C. Hierarchical Autoregressive Joint Decoding

Action probability factorization: Let $\mathbf{w}_t = (w_t^{(1)}, \ldots, w_t^{(q_{i_t})})$ denote the ordered worker generation trajectory and $W_t = \{w_t^{(1)}, \ldots, w_t^{(q_{i_t})}\}$ the resulting team. With trajectory action $\widetilde{a}_t = (i_t, m_t, \mathbf{w}_t)$ and execution action $a_t = (i_t, m_t, W_t)$:

$$\pi_\theta(\widetilde{a}_t \mid s_t) = \pi_O(i_t \mid s_t)\, \pi_M(m_t \mid s_t, i_t) \times \prod_{r=1}^{q_{i_t}} \pi_W\left(w_t^{(r)} \mid s_t, i_t, m_t, \mathbf{w}_t^{(<r)}\right). \tag{18}$$

PPO computes likelihoods on the ordered trajectory $\widetilde{a}_t$, while environment dynamics execute unordered set W_t.

Operation and station pointers: The operation pointer selects from $\mathcal{F}_t^O \subseteq \mathcal{O}^+$, while the station pointer conditions on $\mathbf{h}_{i_t}$ over $\mathcal{F}_{i_t,t}^M$:

$$u_{i,t}^O = \mathbf{a}_O^\top \tanh(\mathbf{W}_g \mathbf{g}_t + \mathbf{W}_O \mathbf{h}_i), \tag{19}$$

$$\pi_O(i \mid s_t) = \exp(u_{i,t}^O) \Big/ \textstyle\sum_{j \in \mathcal{F}_t^O} \exp(u_{j,t}^O), \tag{20}$$

$$u_{m,t}^M = \mathbf{a}_M^\top \tanh(\mathbf{W}_{OM} \mathbf{h}_{i_t} + \mathbf{W}_M \mathbf{h}_m), \tag{21}$$

$$\pi_M(m \mid s_t, i_t) = \exp(u_{m,t}^M) \Big/ \textstyle\sum_{n \in \mathcal{F}_{i_t,t}^M} \exp(u_{n,t}^M). \tag{22}$$

For worker selection, Worker Pointer scores candidates at substep $r \in \{1, \ldots, q_{i_t}\}$:

$$u_{w,t}^{(r)} = \mathbf{v}_W^\top \tanh\left(\mathbf{q}_t^{(r)} + \mathbf{k}_{w,t}^{(r)}\right), \tag{23}$$

$$\pi_W(w \mid s_t, i_t, m_t, \mathbf{w}_t^{(<r)}) = \exp(u_{w,t}^{(r)}) \Big/ \textstyle\sum_{v \in \mathcal{F}_{W,t}^{(r)}} \exp(u_{v,t}^{(r)}). \tag{24}$$

Infeasible candidates receive zero probability. Query $\mathbf{q}_t^{(r)}$ and key $\mathbf{k}_{w,t}^{(r)}$ incorporate skill pressures, dynamic EFT, and marginal scarcity (detailed in Appendix A).

Endogenous worker–station binding: When an initially station-unassigned worker is assigned to an operation at station m_t, the worker becomes dedicated to that station for the remainder of the schedule. The resulting station assignment is retained during subsequent rescheduling:

$$\ell_{w,t+1} = \begin{cases} m_t, & \ell_{w,t} = \varnothing \text{ and } w \in W_t, \\ \ell_{w,t}, & \text{otherwise.} \end{cases} \tag{25}$$

D. PPO Learning and Transfer

With continuation indicator $b_t = (1-d_t)(1-u_t)$, temporal-difference error δ_t and generalized advantage estimator (GAE) $\widehat{A}_t$ [24] are:

$$\delta_t = r_t + \gamma b_t V_{\phi^-}(s_{t+1}) - V_{\phi^-}(s_t), \quad \widehat{A}_t = \delta_t + \gamma\lambda b_t \widehat{A}_{t+1}. \tag{26}$$

Clipped PPO [25] loss with ratio $r_t^{\mathrm{PPO}}(\theta) = \exp(\ell_{t,\theta} - \ell_{t,\theta^-})$ is:

$$\mathcal{L}_{\mathrm{policy}}(\theta) = -\mathbb{E}_t\Big[\min\Big(r_t^{\mathrm{PPO}}(\theta)\widetilde{A}_t, \mathrm{clip}(r_t^{\mathrm{PPO}}(\theta), 1-\epsilon, 1+\epsilon)\widetilde{A}_t\Big)\Big], \tag{27}$$

$$\mathcal{L} = \mathcal{L}_{\mathrm{policy}}(\theta) + c_V \mathbb{E}_t[(V_\phi(s_t) - R_t)^2] - c_H \overline{H}. \tag{28}$$

Shared graph-encoding and pointer parameters accommodate varying numbers of operations and workers without size-specific output heads, with scale-extrapolated schedule quality verified experimentally in Section V. For rescheduling, operation node features expand from 18 to 24 dimensions to incorporate baseline schedules and disturbance signals, widening the first-layer operation projection matrix. Compatible parameters across the graph encoder, pooling layers, and pointer decoders are directly initialized from the trained initial-scheduling model; for the operation projection layer, weights corresponding to existing channels are preserved while new input channels are initialized under standard initialization. The policy is then further trained in the disturbance environment under the composite stability objective $S(\pi^r)$ with inherited worker–station bindings, enabling effective knowledge transfer from baseline scheduling to disturbance recovery.

TABLE IV
REAL FACTORY APAL TEST INSTANCES

Instance	Physical operations	Stations	Role
Medium	283	5	Test
Large	680	5	Test
Industrial-I	2,338	5	Test
Industrial-II	3,182	5	Test

Note: Instance sizes count physical operations only. The source process graphs for Medium, Large, Industrial-I, and Industrial-II additionally contain 7, 35, 64, and 117 structural nodes, respectively (290, 715, 2,402, and 3,299 total nodes).

TABLE V
COMPARISON METHODS, DECISION LOGIC, AND SEARCH BUDGETS

Category	Method	Decision Criteria / Search Budget
Dispatching rules	SPT [26] / LPT	Shortest / longest standard duration first
Dispatching rules	ES / CPM [27] / MSL	Earliest start / critical-path start / min slack
Dispatching rules	Random	Fixed seed random selection
Metaheuristics	Beam Search [28]	Priorities search; init: $w = 4, b = 4$; resched: $w = 2, b = 2$
Metaheuristics	IG [29]	Iterated greedy; init: 160 iters; resched: 3 iters
Metaheuristics	SA [30]	Simulated annealing; init: 120 iters; resched: 3 iters
RL baselines	L2D-PPO-APAL	Graph dispatching policy adapted to APAL
RL baselines	Graph-DDQN-APAL	Graph value-based RL adapted to APAL
Reference	NoReschedule	Negative control: retain unadjusted baseline
Heuristic repair	StabilityAware / CPM	Multi-objective hybrid priority repair

V. EXPERIMENTS AND RESULTS

A. Evaluation Questions and Protocol

Experiments investigate five core evaluation questions:

- **EQ1 (Scale extrapolation)**: Can a locked policy generalize to 283–3,182 physical operations without fine-tuning?
- **EQ2 (Decoding quality)**: Does autoregressive joint decoding outperform heuristics and RL baselines in makespan $H = C_{\max}$?
- **EQ3 (Structural ablations)**: Do heterogeneous graph attention, endogenous binding, and three-tier decisions match APAL physics?
- **EQ4 (Online runtime)**: How does neural forward inference scale against per-instance search algorithms?
- **EQ5 (Disturbance recovery)**: Can the architecture balance feasibility, makespan, and stability under material delays?

Dataset partitioning: The experiments use training, validation, and test sets:

- *Training set:* 24 synthetic variants containing 400–800 physical operations, used for model parameter updates;
- *Validation set:* 6 instances and 54 delay scenarios, used for hyperparameter selection, early stopping, and checkpoint selection;
- *Test set:* 4 real factory instances (Table IV) and 36 fixed disturbance scenarios, reserved for final evaluation after model selection and without further tuning.

The test set comprises four instances derived from real factory APAL process data: Medium (283 physical operations), Large (680), Industrial-I (2,338), and Industrial-II (3,182). All four instances contain five physical stations. Completeness ($R_{\text{complete}} = 1$) and physical validity ($R_{\text{valid}} = 1$) are strict prerequisites for quality ranking. Physical precedence relations are derived by bypassing structural nodes while preserving the dependencies between physical operations.

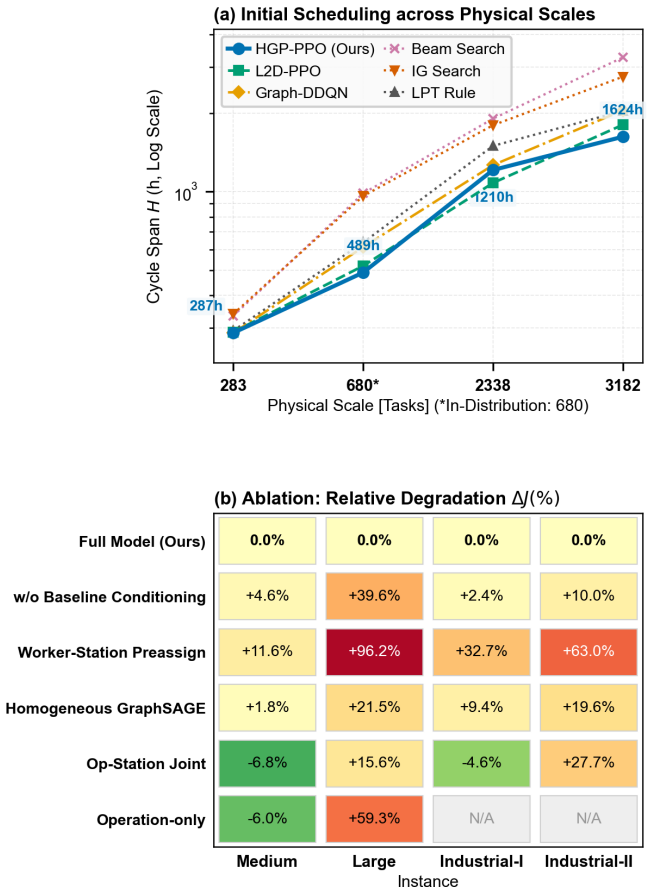


Fig. 3. Initial scheduling performance across physical scales and architectural ablations. (a) Cycle spans H on logarithmic axis across scales. (b) Relative degradation matrix $\Delta J(\%)$ against the full HGP-PPO architecture.

B. Comparison Methods and Statistical Inference

Table V outlines the comparison methods. Initial scheduling is evaluated on deterministic makespan $H = C_{\max}$ (Eq. (2)), supplemented by five random seeds. Rescheduling is evaluated via composite score S (Eq. (13)) across 36 paired disturbance scenarios. Overall significance is assessed via Friedman test across 19 feasible methods ($\chi_F^2(18) = 369.20, p < 0.001$), paired two-sided Wilcoxon signed-rank tests with Holm–Bonferroni correction, and 10,000 stratified bootstrap resamples for 95

C. Initial Scheduling Results Across Scales

Table VI reports initial makespans. Across the four scales, HGP-PPO achieves the lowest equal-weight average makespan of 902.40 h, outperforming L2D-PPO-APAL (922.94 h) by 2.22% and Graph-DDQN-APAL (1053.11 h) by 14.31%. On the largest 3,182-operation instance, HGP-PPO achieves 1,623.69 h, reducing makespan by 181.02 h (10.03

D. Decision Architecture and Configuration Comparisons

Table VII and Fig. 3 analyze structural ablations:

1) **Worker–Station preassignment**: Pre-fixing worker–station assignments inflates average makespan to

TABLE VI
INITIAL SCHEDULING MAKESPAN ACROSS FOUR APAL INSTANCES (H; LOWER IS BETTER)

Category	Method	Medium	Large	Industrial-I	Industrial-II	Mean Across Scales
Graph RL	**Proposed HGP-PPO**	286.96	**489.00**	1209.96	**1623.69**	**902.40**
Allocation ablation	Worker–Station Preassignment	320.31	959.63	1605.13	2645.84	1382.73
RL baselines	L2D-PPO-APAL	288.97	518.65	**1079.44**	1804.71	922.94
RL baselines	Graph-DDQN-APAL	**286.77**	612.29	1263.66	2049.70	1053.11
Metaheuristics	Iterated Greedy (IG)	339.05	961.10	1797.57	2759.01	1464.18
Metaheuristics	Beam Search	333.43	984.30	1904.88	3272.54	1623.79
Metaheuristics	Simulated Annealing (SA)	311.78	1238.00	2417.03	3256.29	1805.77
Dispatching rules	LPT	292.63	637.56	1497.35	2039.50	1116.76
Dispatching rules	Random	310.62	872.52	1414.20	2170.15	1191.87
Dispatching rules	SPT	411.57	1702.77	2479.07	3346.37	1984.94
Dispatching rules	MSL	412.56	1608.50	2496.05	3455.45	1993.14
Dispatching rules	ES-priority	435.13	1773.41	2780.24	3729.75	2179.63
Dispatching rules	CPM	387.28	1597.92	2522.96	3430.06	1984.55

Note: Deterministic masked argmax decoding. All methods share identical forward allocation and physical constraints.

TABLE VII
INITIAL MAKESPANS AND RESCHEDULING FEASIBILITY FOR STRUCTURAL VARIANTS

Method	Structural Modification	Medium	Large	Industrial-I	Industrial-II	Mean	Init Feas.	Resched Feas.
Proposed HGP-PPO	Full heterogeneous graph; joint Op–St–Wk decisions	286.96	489.00	1209.96	1623.69	**902.40**	4/4	**100%**
Homogeneous GraphSAGE	Homogeneous graph encoding without typed message passing	292.02	594.04	1323.22	1941.29	1037.64	4/4	100%
Op–Station Decisions	Rule-based worker selection without explicit worker pointer	**267.40**	565.26	**1154.20**	2073.34	1015.05	4/4	0%
Op-Only Decisions	Heuristic station and worker dispatching	269.66	779.13	Infeas.	Infeas.	—	2/4	100%

Note: Feasibility rates computed across 36 paired disturbance scenarios. Infeasible instances are excluded from mean makespan calculations.

1,382.73 h (+53.2%), indicating that static allocation bottlenecks task execution.

2) **Homogeneous GraphSAGE**: Collapsing heterogeneous relations increases mean makespan to 1,037.64 h (+15.0%).
3) **Op–Station decisions**: Bypassing worker pointers yields low initial makespans on smaller instances but leads to a 0% feasibility rate during rescheduling due to team allocation deadlocks.
4) **Op-only decisions**: Omitting station pointers results in infeasibility on instances with 2,338 and 3,182 operations.

E. Rescheduling Results Under Material Delays

Table VIII summarizes results across 36 scenarios. HGP-PPO achieves the lowest average composite score of $S = 0.8511$, outperforming StabilityAwareRepair (0.8776), L2D-PPO-APAL (0.9152), and Graph-DDQN-APAL (0.9810). Its median and IQM scores are 0.7415 and 0.7670, with the lowest ARD of 26.60% and the lowest average rank (4.56).

Table IX decomposes the physical components. Compared with StabilityAwareRepair, HGP-PPO reduces average makespan by 17.36 h (wins in 32/36 scenarios, $p_{\mathrm{Holm}} < 0.001$) and reference overrun Δ_{ref} by 15.71 h (wins in 29/36 scenarios, $p_{\mathrm{Holm}} < 0.001$). Start-time deviation drops from 33.88 to 30.85 h. Compared with L2D-PPO-APAL, makespan, overrun, and start-time deviation decrease by 11.84, 12.60, and 14.70 h (-32.3%), respectively.

Table X reports paired statistics against the two closest baselines. Against L2D-PPO-APAL, HGP-PPO reduces S by 7.00%, with the 95% bootstrap CI $[-0.1344, -0.0023]$ lying entirely below zero. Against StabilityAwareRepair, S decreases by 3.01% (22 wins vs. 14 losses). Fig. 4 illustrates scenario-level paired differences and win/loss distributions.

F. Online Runtime and Scalability

Table XI and Fig. 5 compare end-to-end online rescheduling runtimes. HGP-PPO executes forward inference in 8.02–123.60 s across the four scales. On Industrial-II (3,182 operations), per-instance heuristic search requires 3,900–4,300 s (≈65–72 minutes), whereas HGP-PPO finishes full-line plan repair in approximately 2 minutes (123.60 s), providing orders-of-magnitude acceleration for real-time shop-floor decision-making.

VI. CONCLUSIONS AND LIMITATIONS

This paper has presented HGP-PPO, a heterogeneous graph reinforcement learning framework for joint scheduling and workforce allocation on aircraft pulsating assembly lines. The hierarchical operation–station–worker autoregressive pointer decoder and dynamic masks avoid explicit enumeration of complete teams, while endogenous worker–station binding coordinates workforce deployment with station workloads. For initial scheduling, a single locked policy is directly evaluated across 283 to 3,182 physical operations without instance-specific fine-tuning, achieving strong scale extrapolation and the lowest average makespan of 902.40 h. For rescheduling, the policy trained under the disturbance environment is evaluated across 36 locked material delay scenarios. All generated schedules in the designated test instances and scenarios pass model constraint validation, achieve a lower average composite recovery score with observable trade-offs among individual stability components, and require an average end-to-end plan repair runtime of approximately 2 minutes (123.60 s) per scenario on Industrial-II.

Limitations and future work: The formulation addresses cyclic operation templates and their plan-level repair. Proposition 1 establishes physical feasibility for the steady-state unfolding of a fixed template; it does not establish feasibility of

TABLE VIII
COMPOSITE SCORES AND MEAN MAKESPANS FOR RESCHEDULING ACROSS 36 MATERIAL DELAY SCENARIOS

Category	Method	Medium	Large	Industrial-I	Industrial-II	Mean S	Makespan (h)	Feasibility
Graph RL	**Proposed HGP-PPO**	0.9572 / 297.7	0.7621 / 674.9	0.8627 / 1223.9	0.8224 / 1709.5	**0.8511**	976.5	**100%**
Model variant	Ablated model without baseline conditioning	0.9700 / 300.1	0.8242 / 682.5	0.9186 / 1239.2	1.0588 / 1786.6	0.9429	1002.1	100%
Ablation	Homogeneous GraphSAGE	1.2236 / 318.9	1.0447 / 682.4	1.0310 / 1263.2	1.1599 / 1806.8	1.1148	1017.8	100%
Ablation	Operation-only decisions	0.9592 / 296.9	1.2479 / 678.6	1.0825 / 1223.8	1.6577 / 1903.3	1.2368	1025.7	100%
RL baselines	L2D-PPO-APAL	0.9439 / 301.3	0.8441 / 663.3	0.8401 / 1223.9	1.0325 / 1764.9	0.9152	988.3	100%
RL baselines	Graph-DDQN-APAL	1.0414 / 306.6	1.0714 / 672.8	0.9394 / 1223.9	0.8716 / 1709.9	0.9810	978.3	100%
Metaheuristics	BeamSearchRepair	1.0575 / 322.9	1.0763 / 752.4	1.2596 / 1403.6	1.3626 / 1987.2	1.1890	1116.5	100%
Metaheuristics	IteratedGreedyRepair	1.0312 / 319.1	1.6770 / 783.1	1.6892 / 1454.1	1.3655 / 1987.5	1.4407	1135.9	100%
Metaheuristics	SimulatedAnnealingRepair	1.1431 / 323.6	2.2088 / 814.1	1.8923 / 1484.4	1.4394 / 2001.8	1.6709	1156.0	100%
Heuristic repair	HybridCPMStabilityRepair	1.1337 / 294.2	1.0463 / 675.4	0.9952 / 1223.9	0.7560 / 1665.9	0.9828	**964.9**	100%
Heuristic repair	StabilityAwareRepair	0.9913 / 310.4	0.7554 / 682.7	0.8045 / 1229.4	0.9591 / 1752.8	0.8776	993.8	100%
Heuristic repair	TaktAwareRepair	0.9103 / 323.8	1.0759 / 775.7	1.0398 / 1307.4	0.9073 / 1816.4	0.9833	1055.8	100%
Heuristic repair	ReleaseAwareRepair	0.7844 / 316.8	1.4584 / 770.5	1.1644 / 1369.4	1.1556 / 1901.5	1.1407	1089.5	100%
Heuristic repair	BottleneckSkillRepair	0.9667 / 326.4	1.4732 / 768.0	1.4282 / 1404.7	0.8398 / 1827.5	1.1770	1081.7	100%
Heuristic repair	SPTRepair	1.0188 / 333.3	0.9794 / 758.4	1.1850 / 1373.5	1.2038 / 1914.2	1.0967	1094.9	100%
Heuristic repair	FullRescheduleCPM	1.3579 / 332.0	1.5415 / 778.9	1.5528 / 1384.3	1.1864 / 1875.6	1.4097	1092.7	100%
Heuristic repair	CPMRepair	1.3513 / 331.1	1.5147 / 773.9	1.5968 / 1396.1	1.1838 / 1880.2	1.4116	1095.3	100%
Heuristic repair	LPTRepair	1.5492 / 345.0	2.6828 / 836.3	1.9494 / 1456.0	1.3546 / 1922.1	1.8840	1139.9	100%
Heuristic repair	RandomRepair	1.5427 / 351.7	3.2131 / 943.2	2.6982 / 1673.2	1.9854 / 2153.4	2.3599	1280.4	100%
Negative control	NoReschedule	—	—	—	—	N/A	N/A	0%

Note: Entries formatted as "S / makespan (h)". Scores normalized by T^0_{ref} and averaged equally across scenarios and instances.

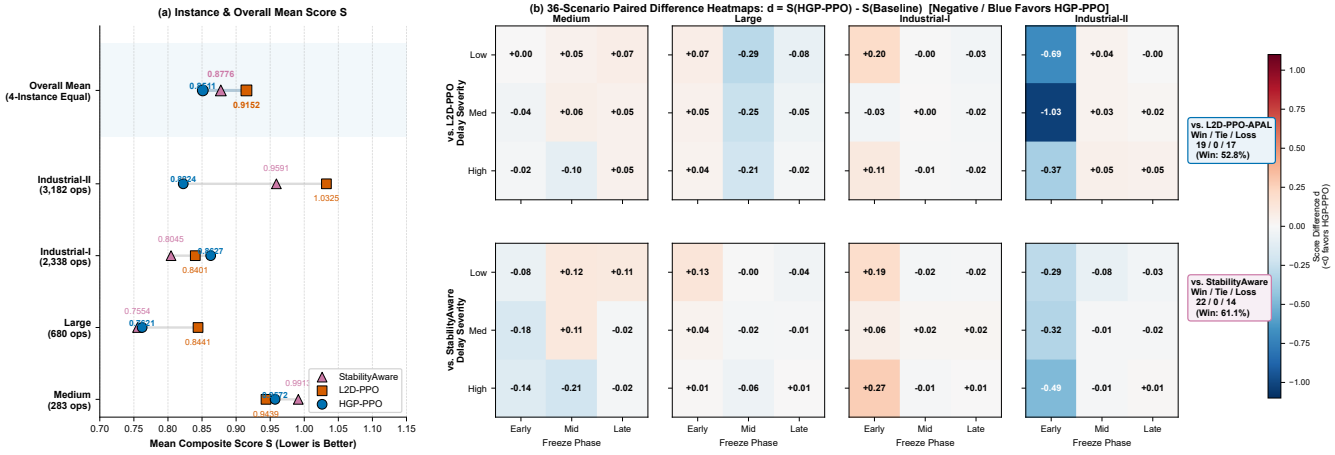


Fig. 4. Paired-difference heatmaps for 36 template rescheduling scenarios under dynamic material delays and instance-level mean recovery scores. (a) Composite recovery scores S for the four physical instance sizes and their equal-weight mean; lower is better. (b) Symmetric diverging heatmaps of paired score differences $d = S_{\text{HGP}} - S_{\text{Baseline}}$ between the proposed method and L2D-PPO-APAL or StabilityAwareRepair across 36 disturbance scenarios (four instances $\times$ three delay levels $\times$ three freeze timings). Blue indicates an advantage for the proposed method, and red indicates an advantage for the baseline. Win / Tie / Loss badges on the right margin summarize the pairwise comparisons.

TABLE IX
RAW RESCHEDULING PHYSICAL COMPONENT DECOMPOSITION

Method	$C_{\max}$ (h)	Δ_{ref} (h)	$\bar{D}_{\text{start}}$ (h)	R_{sta} (%)	R_{team} (%)	Score S
Proposed	976.49	29.51	**30.85**	2.12	95.60	**0.851100**
L2D-PPO	988.33	42.11	45.54	1.93	95.87	0.915151
Graph-DDQN	978.32	29.63	56.31	2.02	96.05	0.980965
BeamSearch	1116.52	167.19	108.85	**0.54**	**0.70**	1.188985
HybridCPM	**964.87**	**18.79**	61.76	1.85	93.53	0.982793
StabilityAware	993.85	45.22	33.88	1.73	94.76	0.877564

Note: R_{team} is evaluated on reschedulable tasks: replacing any member counts as a team modification without implying cross-station mobility.

the transient transition between templates. The model does not explicitly simulate cross-aircraft work-in-process (WIP) movements or buffer congestion during this transition. Updated material release times are assumed known at rescheduling, and standard processing times are deterministic. Future work will extend the framework to transient line simulation, uncertain material arrivals and processing times, and worker fatigue and time-varying productivity.

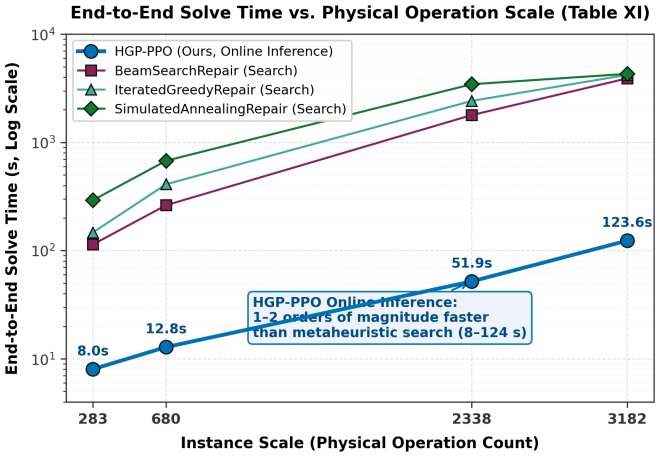


Fig. 5. End-to-end online rescheduling runtime and computational scalability across physical scales on a logarithmic axis.

APPENDIX A
WORKER POINTER ARCHITECTURE AND DYNAMIC FEATURES

Network layers and partial-team pooling: For the selected member sequence $\mathbf{w}_t^{(<r)}$, each member is projected

TABLE X
PAIRED RESCHEDULING STATISTICS AGAINST THE TWO CLOSEST BASELINES

Baseline Method	Mean S Diff	Reduction	Wins / Ties / Losses	p_{Holm}	r_{rb}	Stratified Bootstrap 95% CI
L2D-PPO-APAL	-0.0640	7.00%	19 / 0 / 17	0.7237	-0.069	$[-0.1344, -0.0023]$
StabilityAwareRepair	-0.0265	3.01%	22 / 0 / 14	0.5498	-0.210	$[-0.0652, \ 0.0110]$

Note: Negative differences indicate proposed method superiority. p_{Holm} denotes two-sided Wilcoxon corrected across 18 baselines; CI based on 10,000 resamples.

TABLE XI
END-TO-END ONLINE RESCHEDULING RUNTIME PER SCENARIO (S)

Method	Medium	Large	Industrial-I	Industrial-II
Proposed HGP-PPO	**8.02**	**12.84**	**51.91**	**123.60**
BeamSearchRepair	114.02	262.72	1787.00	3902.89
IteratedGreedyRepair	147.17	410.42	2416.80	4202.01
SimulatedAnnealingRepair	292.42	676.78	3445.39	4296.64

Note: Single-scenario inference runtime excluding offline training and model loading.

via $\mathbf{m}_v = \mathbf{W}_{\mathrm{mem}}\mathbf{h}_v$. To achieve strict permutation invariance, symmetric Sum, Mean, and Max pooling vectors are computed as $\mathbf{p}^{(r)}_{\mathrm{sum},t} = \sum_{v\in\mathbf{w}_t^{(<r)}} \mathbf{m}_v$, $\mathbf{p}^{(r)}_{\mathrm{mean},t} = \frac{1}{\max(1,r-1)} \sum_{v\in\mathbf{w}_t^{(<r)}} \mathbf{m}_v$, and $\mathbf{p}^{(r)}_{\mathrm{max},t} = \max_{v\in\mathbf{w}_t^{(<r)}} \mathbf{m}_v$. Partial-team state $\mathbf{h}^{(r)}_{\mathrm{team},t}$ is then formed by:

$$\mathbf{h}^{(r)}_{\mathrm{team},t} = \mathrm{LeakyReLU}\left(\mathbf{W}_{\mathrm{team}}\left[\mathbf{p}^{(r)}_{\mathrm{sum},t} \,\|\, \mathbf{p}^{(r)}_{\mathrm{mean},t} \,\|\, \mathbf{p}^{(r)}_{\mathrm{max},t}\right]\right). \tag{29}$$

For the initial member selection sub-step ($r = 1$ where $\mathbf{w}_t^{(<1)} = \emptyset$), the pooling outputs are defined as zero vectors $\mathbf{0} \in \mathbb{R}^d$, reducing the partial-team state to $\mathbf{h}^{(1)}_{\mathrm{team},t} = \mathbf{0}$.

Query $\mathbf{q}^{(r)}_t$ projects the concatenation of operation embedding, station embedding, global context, team state, macro pressure, skill consumption ratio, and selection progress:

$$\begin{aligned}\mathbf{q}^{(r)}_t = \mathbf{W}_q\Big[&\mathbf{h}_{i_t} \,\|\, \mathbf{h}_{m_t} \,\|\, \mathbf{g}_t \,\|\, \mathbf{h}^{(r)}_{\mathrm{team},t} \\ &\|\, \mathbf{P}_t \,\|\, \mathbf{c}^{(r)}_t \,\|\, \mathbf{prog}^{(r)}_t\Big] + \mathbf{b}_q,\end{aligned} \tag{30}$$

where global context $\mathbf{g}_t = [\mathbf{g}_M \,\|\, \mathbf{g}_O \,\|\, \mathbf{g}_W] \in \mathbb{R}^{3d}$ strictly maintains the concatenation order of Section IV-B; macro skill pressure is $\mathbf{P}_t = [\mathbf{P}^{\mathrm{all}}_t \,\|\, \mathbf{P}^{\mathrm{near}}_t] \in \mathbb{R}^{10}$; skill consumption ratio is $\mathbf{c}^{(r)}_t = \mathbf{C}^{(r)}_t / \max(\mathbf{S}^{\mathrm{all}}_t, \varepsilon_s) \in [0,1]^5$ with cumulative committed headcount $\mathbf{C}^{(r)}_t = \sum_{v\in\mathbf{w}_t^{(<r)}} \mathbf{H}_v$; and selection progress is $\mathbf{prog}^{(r)}_t = \left[\frac{r-1}{q_{i_t}}, \frac{q_{i_t}-(r-1)}{q_{i_t}}\right]^\top \in [0,1]^2$.

The candidate key combines candidate embedding, skill pressure exposure, and dynamic features:

$$\mathbf{k}^{(r)}_{w,t} = \mathbf{W}_k[\mathbf{h}_w \,\|\, \mathbf{e}_{w,t}] + \mathbf{W}_{\mathrm{eft}}\mathbf{f}^{\mathrm{eft},(r)}_{w,t} + \mathbf{W}_{\mathrm{sc}}\mu^{(r)}_{w,t}, \tag{31}$$

where skill pressure exposure $\mathbf{e}_{w,t} = [\mathbf{H}_w \odot \mathbf{P}^{\mathrm{all}}_t \,\|\, \mathbf{H}_w \odot \mathbf{P}^{\mathrm{near}}_t] \in \mathbb{R}^{10}$ remains fixed across selection sub-steps $r = 1, \ldots, q_{i_t}$ for operation i_t. Pointer scoring evaluates $\mathbf{v}_W^\top \tanh(\mathbf{q}^{(r)}_t + \mathbf{k}^{(r)}_{w,t})$, masked by -10^4 on invalid candidates.

Global and near-term skill pressure: Let $R_{jk} \in \{0,1\}$ indicate whether operation j requires skill k, $H_{wk} \in \{0,1\}$ indicate whether worker w possesses skill k, and $\widetilde{p}_j$ be the normalized standard processing time. Let $\mathcal{U}_t \subseteq \mathcal{O}^+$ denote unscheduled physical operations, and let $\mathcal{R}_t \subseteq \mathcal{U}_t$ denote ready operations. Let $\widetilde{a}_{w,t} = \exp(x^{\log_\mathrm{wait}}_{w,t}) - 1 \geq 0$ denote worker normalized wait time, and let $\mathcal{W}^q_t$ denote available workers:

$$D^{\mathrm{all}}_{k,t} = \sum_{j\in\mathcal{U}_t} R_{jk}\widetilde{p}_j q_j, \quad D^{\mathrm{near}}_{k,t} = \sum_{j\in\mathcal{R}_t} R_{jk}\widetilde{p}_j q_j, \tag{32}$$

$$S^{\mathrm{all}}_{k,t} = \sum_{w\in\mathcal{W}} H_{wk}, \quad S^{\mathrm{near}}_{k,t} = \sum_{w\in\mathcal{W}^q_t} H_{wk}\exp(-\widetilde{a}_{w,t}/\tau_p), \tag{33}$$

$$P^h_{k,t} = \log\left(1 + D^h_{k,t}/\max(S^h_{k,t}, \varepsilon_s)\right), \quad h \in \{\mathrm{all}, \mathrm{near}\}. \tag{34}$$

Wait discounting temperature is set to $\tau_p = 1.0$, and smoothing constant $\varepsilon_s = 10^{-6}$ prevents division by zero.

Dynamic EFT: With partial team wait $A^{(r-1)}_t$, cumulative efficiency $E^{(r-1)}_t$, station slot wait $\widetilde{a}_{m_t,t}$, and candidate baseline efficiency η_w, tentative completion time is:

$$\widehat{C}^{(r)}_{w,t} = \max\left(A^{(r-1)}_t, \widetilde{a}_{w,t}, \widetilde{a}_{m_t,t}\right) + \frac{\widetilde{p}_{i_t} q_{i_t}}{\rho^{r-1}(E^{(r-1)}_t + \eta_w)}, \tag{35}$$

with team synergy discount $\rho = 0.95$. Across feasible candidate set $\mathcal{F}^{(r)}_{W,t}$, compute minimum completion $C^{(r)}_{\min,t} = \min_{u\in\mathcal{F}^{(r)}_{W,t}} \widehat{C}^{(r)}_{u,t}$, mean $\mu^{(r)}_{C,t} = \frac{1}{|\mathcal{F}^{(r)}_{W,t}|}\sum_{u\in\mathcal{F}^{(r)}_{W,t}} \widehat{C}^{(r)}_{u,t}$, and population standard deviation $\sigma^{(r)}_{C,t}$; the dynamic EFT feature consists of relative gap and clipped z-score:

$$\sigma^{(r)}_{C,t} = \sqrt{\frac{1}{|\mathcal{F}^{(r)}_{W,t}|}\sum_{u\in\mathcal{F}^{(r)}_{W,t}} \left(\widehat{C}^{(r)}_{u,t} - \mu^{(r)}_{C,t}\right)^2}, \tag{36}$$

$$\mathbf{f}^{\mathrm{eft},(r)}_{w,t} = \begin{bmatrix} \mathrm{clip}\left(\frac{\widehat{C}^{(r)}_{w,t} - C^{(r)}_{\min,t}}{\widetilde{p}_{i_t}}, 0, 10\right) \\ \mathrm{clip}\left(\frac{\widehat{C}^{(r)}_{w,t} - \mu^{(r)}_{C,t}}{\max(\sigma^{(r)}_{C,t}, 10^{-6})}, -10, 10\right) \end{bmatrix}. \tag{37}$$

When $|\mathcal{F}^{(r)}_{W,t}| = 1$, the standard deviation is zero, and denominator clamp 10^{-6} ensures numerical safety.

Marginal skill scarcity: Define the skill pressure function as:

$$P^{(r)}_{k,t}(x) = \log\left(1 + \frac{D^{\mathrm{all}}_{k,t}}{\max\left(S^{\mathrm{all}}_{k,t} - C^{(r)}_{k,t} - x, \ \varepsilon_s\right)}\right), \tag{38}$$

and the marginal pressure increment as:

$$\Delta P^{(r)}_{k,t}(w) = P^{(r)}_{k,t}(H_{wk}) - P^{(r)}_{k,t}(0). \tag{39}$$

Candidate worker w's marginal skill scarcity is then:

$$\mu^{(r)}_{w,t} = \sum_{k\in\mathcal{K}} H_{wk}(1 - R_{i_t k})\max\left(0, \ \Delta P^{(r)}_{k,t}(w)\right), \tag{40}$$

TABLE XII
KEY REPRODUCIBILITY HYPERPARAMETERS

Category	Parameter	Value / Configuration
Instances	Training / Validation / Test	24 / 6 / 4 instances
Disturbances	Validation / Test	54 / 36 (9 scenarios per instance)
Network	Architecture	5 GATv2 layers, 4 heads, $d = 128$
Model size	Trainable parameters	$\approx$11.2 M
PPO	Learning rate / γ / GAE λ	1.0×10^{-4} / 0.999 / 0.9
PPO	Epochs / clip range ϵ	2 / 0.20 $\rightarrow$ 0.10
Optimization	Optimizer / weight decay	ScheduleFree AdamW / 1.0×10^{-4}
Gradients	Actor / Critic clip norms	0.5 / 0.05
Batching	Batch size / environments	64 or 128 / 4 envs, grad accum 4
Budget	Initial / rescheduling epochs	60 epochs / 120 epochs (two stages)

where $C_{k,t}^{(r)} = \sum_{v \in \mathbf{w}_t^{(<r)}} H_{vk}$ is the committed skill k headcount across prior $r-1$ sub-steps. The factor $(1 - R_{i_t k})$ excludes skills required by operation i_t, penalizing unrequired skill capacity drain and mitigating multiskilled worker misallocation.

APPENDIX B
REPRODUCIBILITY CONFIGURATION AND METRIC DEFINITIONS

Table XII provides the key reproducibility hyperparameters. **Metric definitions**: 1) *IQM*: Arithmetic mean of middle 50% scores after 25% trimming. 2) *ARD*: For method m over scenario set $\mathcal{S}$ ($|\mathcal{S}| = 36$): $\mathrm{ARD}_m = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \frac{S_{ms} - S_s^*}{S_s^*} \times 100\%$, where $S_s^* = \min_k S_{ks}$ is the best score observed across evaluated methods in scenario s. 3) *Rank-biserial correlation (r_{rb})*: Wilcoxon effect size $r_{\mathrm{rb}} = (W^+ - W^-)/(W^+ + W^-) \in [-1, 1]$.

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